

Semiconductor Device Modelling

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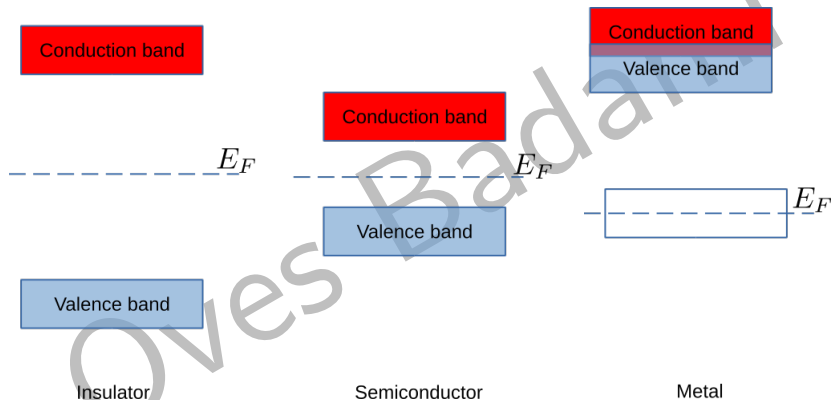
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Semiconductor Fundamentals

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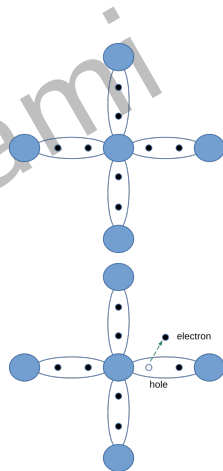
Material Classification – Conductivity



- Semiconductor are the materials that have their conductivity in between the metals and insulators
- Bandgap of the semiconductor also lies between metal (have zero bandgap) and insulators.
- Major classification of semiconductors
 - Elemental semiconductor examples Si, Ge, α -Sn
 - Compound semiconductors
 - III-V – examples InAs, GaAs
 - II-VI – examples ZnS, CdSe
- Elemental semiconductor – Bonds are covalent in nature
- Compound semiconductors – Bonds are covalent+ionic in nature

Bond Picture – Intrinsic Semiconductor

- In a pure elemental semiconductor the core is considered have $+4$ charge.
- At 0 K all the electrons are bonded.
- At higher temperatures the atoms vibrate around their mean position.
- Due to these vibrations the bounded electrons can get pushed out of the bonds there by becoming free. This process is called as *generation*.
- This leaves a vacancy behind called as *hole*.
- Neighbouring electrons jump into it and thus leaving a vacancy in another location thus giving an impression that *hole* is moving.

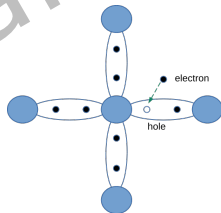


Bond Picture – Intrinsic Semiconductor

- When electron while moving around may bump in to a hole and may annihilate each other. This process is called as *recombination*.
- Due to the generation process certain number of electron and hole pairs are created.

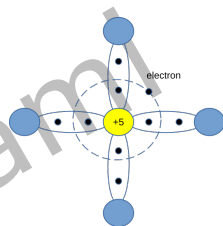
$$n = p = n_i$$

- In thermal equilibrium the carrier density remains constant due to the recombination and generation process.
- The value of $n_i (\propto e^{-\frac{E_g}{2k_B T}})$ depends strongly on the band gap of the semiconductor.

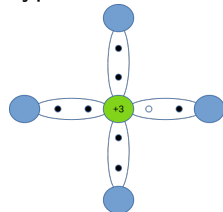


Bond Picture – Extrinsic Semiconductor

- When we add impurities to an intrinsic semiconductor it is called as an extrinsic semiconductor.
- For Si, boron, phosphorus, arsenic are principle impurities which are intentionally added to a pure Si.
 - The energy level of the dopant atom is close to conduction/valence band so that it is ionized at room temperature.
- When an atom from a group V (Donor atom) is added to an intrinsic semiconductor, the resultant semiconductor is called a n-type. ($n \gg p$)
- When an atom from a group III (Acceptor atom) is added to an intrinsic semiconductor, the resultant semiconductor is called a p-type. ($p \gg n$)



n-type semiconductor



p-type semiconductor

Extrinsic Semiconductor – Hydrogen model

- In a hydrogen atom the radius of the orbit of the electron is given by

$$r_B = \frac{4\pi\epsilon_0\hbar^2}{m_0q^2}$$

- Using this as an approximate result the radius of the electron associated with a dopant in the silicon is

$$r = \frac{4\pi\epsilon_0\epsilon_R\hbar^2}{m_0m^*q^2} = r_B\epsilon_R\frac{m_0}{m^*}$$

m^* is the conductivity effective mass

- For silicon it $m^* = 0.26m_0$ and $\epsilon_R = 11.7$.
- Using these values the $r \approx 2.4 \text{ nm}$. Note that the bond length is Silicon is 0.35 nm .

Extrinsic Semiconductor – Hydrogen model

- The ionization energy for the electron in the lowest energy level in a hydrogen atom is given by

$$E = -\frac{m_0 q^4}{2\hbar^2 (4\pi\epsilon_0)^2} = -13.6 \text{ eV}$$

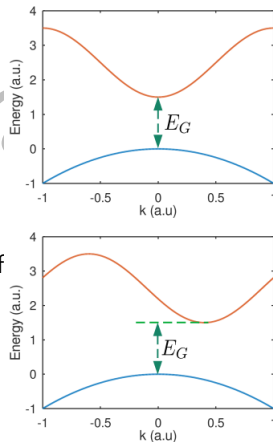
- eV is the unit of energy and conversion is given by J/q
- Considering the dopant and its electron a similar to the hydrogen atom and using the values of the silicon

$$E = -\frac{m_0 m^* q^4}{2\hbar^2 (4\pi\epsilon_0 \epsilon_R)^2} \approx 25 \text{ meV}$$

- This the energy that it takes for an electron to get free from the influence of its parent atom (i.e. make transition from donor energy level to the conduction band).
- The thermal energy at room temperature is $k_B T \approx 26 \text{ meV}$

Band Picture – Semiconductor

- Direct bandgap semiconductor
 - Momentum conservation.
 - Example - InAs
- Indirect bandgap semiconductor
 - Necessity of a third particle for conservation of momentum – phonon.
 - Example - Silicon, Germanium



Band Picture – Extrinsic Semiconductor

- Generally the carriers in the semiconductor essentially remain at the bottom of the band because of scattering.
 - E_C represents the bottom of the conduction band.
 - E_V is the top of the valence band.
 - E_i is the intrinsic Fermi level (i.e. Fermi energy for an intrinsic semiconductor).
 - E_F is the Fermi energy of the semiconductor.
- The bottom of the band can be represented within the parabolic effective mass framework (i.e. constant effective mass for the entire band)
- In n-type semiconductor the number of free electron concentration (n) is much more than the hole concentration (p).
 - The Fermi energy, (E_F) moves towards the conduction band.

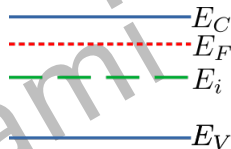


Figure: n-type



Figure: p-type

Carrier Density – n-type

- Let dn be the number of electrons in the conduction band in the interval $(E, E + dE)$

$$dn = N(E)f(E)dE = \frac{4\pi}{h^3} (2m_e^*)^{1.5} (E - E_C)^{0.5} \frac{1}{1 + e^{\left(\frac{E - E_F}{k_B T}\right)}} dE$$

- In a non-degenerate (Fermi-Dirac statistics, $f(E)$, is approximately equivalent to Maxwell-Boltzmann statistics which is equivalent to $E_C - E_F \geq 3k_B T$) n-type semiconductor

$$\begin{aligned} n &= \frac{4\pi}{h^3} (2m_e^*)^{1.5} \int_{E_C}^{\infty} (E - E_C)^{0.5} e^{\left(-\frac{E - E_F}{k_B T}\right)} dE \\ &= 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{1.5} e^{\left(-\frac{E_C - E_F}{k_B T}\right)} = N_C e^{\left(-\frac{E_C - E_F}{k_B T}\right)} \end{aligned}$$

- N_C is the *Effective* density of states and not the *Density of States*

Carrier Density – p-type

- $f(E)$ gives the probability of occupation of electron at an energy E .
- Holes represent an absence of electron so the probability of locating a hole at energy E is $1 - f(E)$.
- Let dp be the number of holes in a band between energy $(E, E + dE)$

$$\begin{aligned} dp &= N(E)(1 - f(E)) dE \\ &= N(E) \frac{1}{1 + e^{\frac{E_F - E}{k_B T}}} dE \end{aligned}$$

- Assuming a non-degenerate semiconductor we get

$$p = N_V e^{\left(-\frac{E_F - E_V}{k_B T}\right)} \quad N_V = 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{1.5}$$

Law of Mass Action

- In an intrinsic semiconductor $E_F = E_i$
- In an intrinsic semiconductor and electron (n_i) and holes (p_i) are created simultaneously. So $n_i = p_i$

$$n_i = N_C e^{\left(-\frac{E_C - E_i}{k_B T}\right)} \quad p_i = N_V e^{\left(-\frac{E_i - E_V}{k_B T}\right)}$$

Extrinsic semiconductor – n-type

- Combining these equation for electron density with the earlier one

$$n = n_i e^{\frac{E_F - E_i}{k_B T}}$$

Extrinsic semiconductor – p-type

- Combining these equation for hole density with the earlier one

$$p = n_i e^{\frac{E_i - E_F}{k_B T}}$$

Law of Mass Action

- Multiplying the above equations we get

$$np = n_i^2 = N_C N_V e^{-\frac{E_C - E_V}{k_B T}} = N_C N_V e^{-\frac{E_g}{k_B T}}$$

Remarks:

$$n = N_C e^{-\frac{E_C - E_F}{k_B T}} = n_i e^{\frac{E_F - E_i}{k_B T}} \quad p = N_V e^{-\frac{E_F - E_V}{k_B T}} = n_i e^{\frac{E_i - E_F}{k_B T}}$$

If $n \gg p$ i.e. it is an n-type semiconductor

- This corresponds to the $E_F - E_i \geq 0$ and E_F lies closer to the conduction band, E_C ,

If $p \gg n$ i.e. it is an p-type semiconductor

- This corresponds to the $E_F - E_i \leq 0$ and E_F lies closer to the valence band, E_V ,

Fermi level position

- A piece of semiconductor is neutral

$$N_D - N_A + p - n = 0$$

- In intrinsic semiconductor the net doping ($N_D - N_A$) is zero

$$N_V e^{-\frac{E_i - E_V}{k_B T}} - N_C e^{-\frac{E_C - E_i}{k_B T}} = 0$$

$$\Rightarrow \frac{E_C + E_V - 2E_i}{k_B T} = \frac{3}{2} \ln \left(\frac{m_e^*}{m_h^*} \right)$$

$$\Rightarrow E_i = E_V + \frac{E_g}{2} - \frac{3}{4} k_B T \ln \left(\frac{m_e^*}{m_h^*} \right)$$

- The Fermi level for an intrinsic semiconductor doesn't lie exactly in the middle of the band gap.

Fermi Level Position – n-type

- In an extrinsic semiconductor, let us say n-type

- $N_A = 0$

- The hole concentration is very small $p = \frac{n_i^2}{n} = \frac{n_i^2}{N_D}$

$$N_C e^{-\frac{E_C - E_F}{k_B T}} = N_D$$
$$\Rightarrow E_F = E_C + k_B T \ln \left(\frac{N_D}{N_C} \right)$$

- Another form for calculating Fermi energy is

$$E_F = E_i + k_B T \ln \left(\frac{N_D}{n_i} \right)$$

Ionization of Impurities

- At low temperatures the electrons (or holes) would not have enough energy to jump from the the dopant energy level to the conduction (or valence) band.
- Simple extrapolation of Fermi-Dirac statistics to calculate the ionized impurity is incorrect.
 - The ionized donor atom can accept an electron with either spin-up or spin-down



Figure: n-type

$$\frac{N_D^+}{N_D} = \frac{1}{1 + g_D e^{\frac{E_F - E_D}{k_B T}}} \quad g_D = 2$$

- The valence band has an additional degeneracy of 2 due to heavy and light hole bands.



Figure: p-type

$$\frac{N_A^-}{N_A} = \frac{1}{1 + g_A e^{\frac{E_A - E_F}{k_B T}}} \quad g_A = 4$$

Carrier Density and Temperature

- From charge neutrality for an n-type semiconductor

$$N_D N_C \frac{1}{N_C + 2n e^{\frac{E_C - E_D}{k_B T}}} + \frac{N_C N_V}{n} e^{-\frac{E_g}{k_B T}} = n$$

- At low temperature – Freeze out range

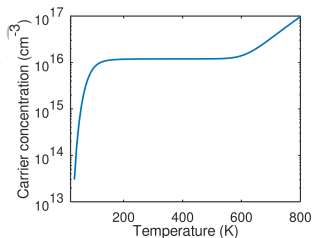
$$n = \left(\frac{N_D N_C}{2} \right)^{0.5} e^{-\frac{E_C - E_D}{2k_B T}}$$

- In the impurity ionization range

$$n = N_D$$

- 2nd term in LHS dominates at high temperature – Intrinsic range.

$$n = (N_C N_V)^{0.5} e^{-\frac{E_g}{2k_B T}}$$

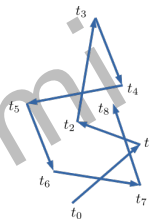


Carrier Transport – Drift Carrier Transport

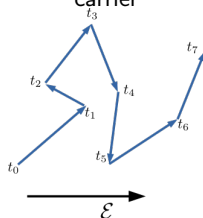
- Electrons are considered as particles.
- At temperature, $T > 0$ the electrons have a finite kinetic energy due to the thermal energy.

$$\frac{1}{2} m_e^* v_{th}^2 = \frac{3}{2} k_B T$$

- The motion of the carrier gets interrupted by scattering.
- The net exchange of energy is zero
- When an electric field, $\mathcal{E}$, is switched-on the random motion of the carrier get a direction. This motion is called drift motion.
 - Motion of a feather in a gravity.
- Between collisions the electrons gain energy from the electric field and give to the atoms in the crystal.



Random motion of a carrier



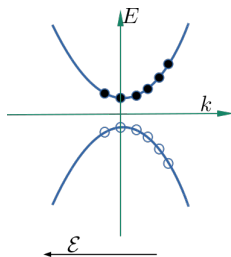
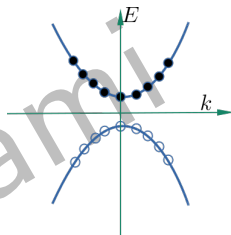
Directed flow of carrier in an electric field $\mathcal{E}$

Hole Transport

- Under equilibrium the net number of electrons having positive momentum is equal to the number of electrons having negative momentum. And hence the current is zero
- Under non-equilibrium conditions, current flows only and only if the band is partially filled.
 - Water bottle analogy
- In a completely filled band

$$j = -q \sum_{i=all} v_i = 0$$

- The bands are symmetric with respect to the momentum.



Hole Transport

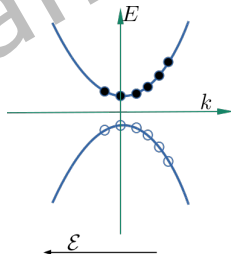
- In the top band

$$j_n = -q \sum_i v_i$$

- In the lower band

$$\begin{aligned} j_p &= -q \sum_i v_i \\ \Rightarrow j_p &= -q \sum_i v_i + q \sum_{j=\text{all}} v_j \\ \Rightarrow j_p &= q \sum_{k=\text{empty}} v_k \end{aligned}$$

- This represents a particle with a charge $+q$. This is what is referred to as hole.
- The electron and hole (as particles) flow in opposite direction.
- Opposite charge results in a net flow.



- Electrons or holes moving under the influence of an electric field cannot continuously accelerate. The scattering of the charge carriers provides a counter balance and thus reaching steady state.
- In a group of N_0 carriers moving in an electric field, $\mathcal{E}$ at time $t = 0$.
- Let $N(t)$ be the number of carriers that haven't undergone scattering at time, t
- Then the rate of decrease of $N(t)$ that haven't undergone scattering is proportional to $N(t)$

$$-\frac{dN(t)}{dt} \propto N(t); \quad -\frac{dN(t)}{dt} = \frac{1}{\tau} N(t) \implies N(t) = N_0 e^{-\frac{t}{\tau}}$$

where τ is the mean free time or time between scattering.

- So if a charge carrier is moving with a momentum p the rate of change of momentum due to a collision will be (in steady state)

$$\frac{dp}{dt} = -\frac{p}{\tau}$$

- In steady state the applied force ($Nq\mathcal{E}$) balances the collisions to result in a net momentum p

$$-\frac{p}{\tau} = Nq\mathcal{E} \implies \langle p \rangle = \frac{p}{N} = -q\tau\mathcal{E}$$

$$\langle \nu \rangle = \frac{\langle p \rangle}{m_n^*} = -q\frac{\tau}{m_n^*}\mathcal{E} = \mu_n\mathcal{E}$$

- μ_n is the constant of proportionality between the drift velocity and electric field.
- $\langle \nu \rangle$ is the drift velocity and is much smaller than the thermal velocity ν_{th} .

Mobility – Scattering Mechanisms

- Scattering occurs only and only when the periodicity of the crystal lattice is disturbed
- Phonon scattering
- Impurity scattering
- Neutral impurity scattering
- Alloy scattering
- Surface Roughness scattering
-

Drift Current

- Current is defined as the rate of flow of charge

$$J_{drift,n} = - (qn) \langle v \rangle$$

$$J_{drift,n} = - (qn) q \frac{\tau}{m_n^*} \mathcal{E} = \frac{q^2 n \tau}{m_n^*} \mathcal{E}$$

$$J_{drift,n} = \sigma_n \mathcal{E} \quad \sigma_n = \frac{q^2 n \tau}{m_n^*}$$

- Thus the drift current can be written as

$$J_{drift,n} = qn\mu_n \mathcal{E}$$

- Similarly for holes as well, the drift current is

$$J_{drift,p} = qp\mu_p \mathcal{E}$$

- The total *drift* current in a semiconductor is thus

$$J_{drift} = J_{drift,n} + J_{drift,p} = q(n\mu_n + p\mu_p) \mathcal{E} = \sigma \mathcal{E}$$

This is the microscopic version of *Ohm's law*

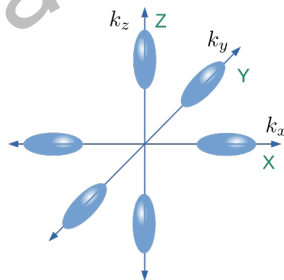
A digression into effective masses

- We have seen until effective masses at 2 different places
 - Density of States calculation, m_{DoS}
 - Carrier flow, m_{Trans}
- m_{DoS} is used for calculating the available states. An example for Si (having 6 equivalent valleys)

$$m_{DoS} = 6 (m_l m_t^2)^{1/3}$$

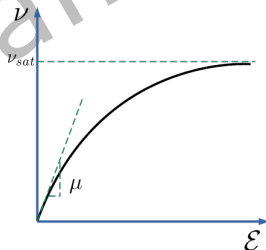
- m_{Trans} is used for calculating the current flow and consequently only certain components are needed.
- For an electron moving along X, Y or Z direction

$$m_{Trans} = 3 \left(\frac{1}{m_l} + \frac{2}{m_t} \right)^{-1}$$



Is mobility really constant ?

- Mobility is a proportionality constant between electric field($\mathcal{E}$) and drift velocity (ν).
- At low field, μ is constant
 - For Si this is more or less valid until ($\mathcal{E} = 10^3 \text{ V/cm}$)
- At high electric field, velocity tends to saturate for most of the semiconductors to thermal velocity.
- Electrons whose drift velocity becomes more or less equal to the thermal velocity are called as *hot electrons*
- Any increase in the carrier energy is transferred to the lattice
 - Inelastic scattering



Mobility versus Temperature

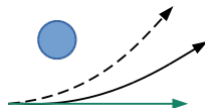
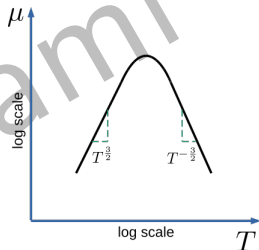
Low temperature

- The vibration of atoms isn't large enough.
- A moving electron gets deflected from its path due to impurity.

High temperature

- Scattering between the the electrons and the vibrating atoms
- Screening of the impurity ions
- Net mobility assuming that all the scattering events are independent – Matthiessen Rule

$$\frac{1}{\mu} = \frac{1}{\mu_1} + \frac{1}{\mu_2} + \dots$$



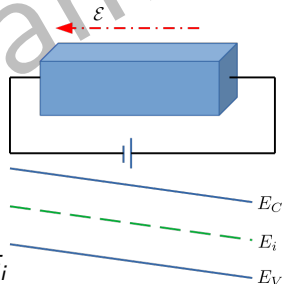
Electric field – Different perspectives

- Let us apply a bias V across a semiconductor.
- Application of positive bias lowers the energy level (or equivalently negative bias lowers the energy level) – Electron energy
- In this case the electric field, $\mathcal{E}$ can be written as

$$\mathcal{E} = -\frac{dV}{dx}$$

- The magnitude slope of the of the potential is same as magnitude of the slope of the E_C or E_i or E_V but in opposite direction.

$$\Rightarrow \mathcal{E} = -\frac{dV}{dx} = \frac{1}{q} \frac{dE_C}{dx} = \frac{1}{q} \frac{dE_V}{dx} = \frac{1}{q} \frac{dE_i}{dx}$$



Diffusion Current

- Diffusion is the process of movement from higher concentration area to lower concentration.

$$J(x) \propto -\frac{dC(x)}{dx} \implies J(x) = -qD \frac{dC(x)}{dx}$$

- The phenomenon of diffusion current is particularly important in the semiconductors.
- Consider that there is a rapid generation of electron and hole pairs in a particular region of a device. These excess carrier would start to move (more correctly diffuse) away from where they are generated.

Diffusion Current – Formal Approach

- Let c_1 be the concentration on the left and c_2 be the carrier concentration on the right of x_0
- No. of particles moving from left to right, ϕ_{l2r}

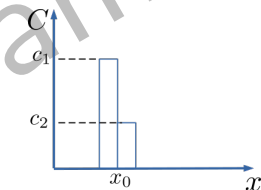
$$\phi_{l2r} = \frac{1}{2} c_1 \bar{l} A$$

- No. of particles moving from right to left, ϕ_{r2l}

$$\phi_{r2l} = \frac{1}{2} c_2 \bar{l} A$$

- Therefore the rate of flow of particle density

$$\phi = \frac{\bar{l}}{2\bar{t}} (c_1 - c_2) = -\frac{\bar{l}^2}{2\bar{t}} \frac{dc}{dx} = -D \frac{dc}{dx}$$



the width of the each sampling is $\bar{l}$

$\bar{l}$ - mean free path; $\bar{t}$ - mean free time

Electron and hole Diffusion current

- The negative sign in the diffusion current stems essentially from the movement of the particles from higher concentration to the lower concentration.
- The diffusion current due to electron and holes

$$J_{n,diff} = (-q) \left(-D_n \frac{dn}{dx} \right) = qD_n \frac{dn}{dx}$$

$$J_{p,diff} = (q) \left(-D_p \frac{dp}{dx} \right) = -qD_p \frac{dp}{dx}$$

Drift and Diffusion current

- Total electron current

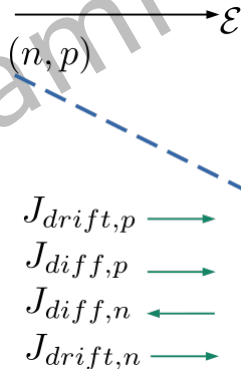
$$J_n = qn\mu_n\mathcal{E} + qD_n\frac{dn}{dx}$$

- Similarly for holes as well

$$J_p = qn\mu_p\mathcal{E} - qD_p\frac{dp}{dx}$$

- Thus the total current, J

$$J = J_n + J_p$$



Einstein's Relationship – Non-degenerate Semiconductor

- In equilibrium condition, by definition

$$J = J_n + J_p = 0$$

- In a non-degenerate n-type semiconductor
- The concentration of holes is very small and can be neglected

$$J_n = J_{diff,n} + J_{drift,n} = qD_n \frac{dn}{dx} + qn\mu_n \mathcal{E} = 0$$
$$n = N_C e^{-\frac{E_C - E_F}{k_B T}} \implies \frac{dn}{dx} = -\frac{1}{k_B T} n \frac{dE_C}{dx} = -\frac{q}{k_B T} n \mathcal{E}$$

the Fermi level in the equilibrium is constant.

- Combining the above equations we get

$$\frac{D_n}{\mu_n} = \frac{k_B T}{q}$$

Quasi-Fermi Level

- The concept of Fermi level is very useful in understanding the carrier statistics and has a very intuitive feel.
- The Fermi level is defined only under equilibrium case and loses all validity in non-equilibrium case.
- We define quasi-fermi levels for electrons and holes which allow to write

$$n = n_i e^{\frac{F_n - E_i}{k_B T}} \quad p = n_i e^{\frac{E_i - F_p}{k_B T}}$$

- For electrons

$$\frac{dn}{dx} = \frac{n_i}{k_B T} e^{\frac{F_n - E_i}{k_B T}} \left(\frac{dF_n}{dx} - \frac{dE_i}{dx} \right) = \frac{n}{k_B T} \left(\frac{dF_n}{dx} - \frac{dE_i}{dx} \right)$$

Quasi-Fermi Level

- Substituting it in the current equations

$$J_n = qn\mu_n\mathcal{E} + qD_n\frac{n}{k_B T} \left(\frac{dF_n}{dx} - q\mathcal{E} \right) = n\mu_n\frac{dF_n}{dx} = \frac{\sigma_n}{q} \frac{dF_n}{dx}$$

where we have employed Einstein's relationship and $\mathcal{E} = \frac{1}{q} \frac{dE_i}{dx}$

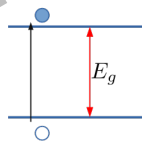
- Similarly for holes as well

$$J_p = p\mu_p\frac{dF_p}{dx} = \frac{\sigma_p}{q} \frac{dF_p}{dx}$$

- So when we have a system that has been driven out of its equilibrium state (i.e. we have electrons/holes/both) more than their equilibrium value how does the system get back ? What drives the system to its equilibrium value.

Excess carrier life time

- When we bombard a semiconductor with photons of energy greater than the bandgap, electron-hole pairs are generated. This is just one of the ways to drive the system out of the equilibrium.
- This is a non-equilibrium scenario.
- Let the number generated electrons (holes) be Δn (Δp).
- so the total electron (hole) concentration is $n = n_0 + \Delta n$ ($p = p_0 + \Delta p$)
- Low level injection: $\Delta n \ll n_0$ and $\Delta p \ll p_0$
- High level injection: Excess carrier are comparable or more than the equilibrium density



Excess carrier life time

- In equilibrium the generation rate is exactly balanced by the recombination rate
 - This results in constant carrier concentration, n_i
- When the light is shines on the semiconductor, the generation rate increases and a steady state is reached
 - Due to the increase in the concentration of the carriers it is reasonable that the recombination rate also increases beyond its thermal rate.

$$G_{th} + G_L - R = 0$$

- After the steady rate the light is switched-off, so the net recombination rate is

$$R_{net} = -\frac{dp}{dt} = -\frac{dn}{dt} = R - G_{th} = \alpha_R (np - n_i^2)$$

assuming $R \propto np$ and α_R is a temperature dependent constant

Excess carrier life time

- Assuming low level injection.

$$pn - n_i^2 = (p_0 + \Delta p)(n_0 + \Delta n) - n_i^2 \approx p_0 \Delta n + n_0 \Delta p$$

- During the generation process $\Delta p = \Delta n$
- Consider an n-type silicon with doping N_D
- The equilibrium electron concentration $n_0 = n_{n0} = N_D$ and equilibrium hole concentration is $p_0 = p_{n0} = \frac{n_i^2}{N_D}$

$$R_{net} = \alpha_R (n_{n0} + p_{n0}) \Delta p \approx \alpha_R N_D \Delta p = \frac{1}{\tau_p} \Delta p$$

$$\frac{dp_n}{dt} = \frac{d\Delta p}{dt} = -R_{net} = -\frac{\Delta p}{\tau_p} \implies \Delta p(t) = \Delta p(0) e^{-\frac{t}{\tau_p}}$$

$$\implies p_n(t) = p_{n0} + (p_n(0) - p_{n0}) e^{-\frac{t}{\tau_p}}$$

τ_p is the excess carrier life time. Similar relationship can also be worked out for electrons in a p-type semiconductor.

Direct Band to Band Recombination

- Radiative Recombination, R_R ,

$$R_R = \alpha np$$

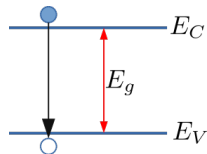
- In thermal equilibrium the radiative recombination must be balanced by the radiative generation process

$$\Rightarrow G_{th} = \alpha n_i^2$$

- The net recombination rate, R_{net} is given by

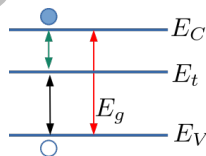
$$R_{net} = \alpha (n_0 + p_0) \Delta p = \frac{G_{th}}{n_i^2} (n_0 + p_0) \Delta p = \frac{1}{\tau_R} \Delta p$$

- The radiative recombination isn't the dominant mechanism in indirect semiconductor (like Si, Ge)



Indirect Recombination

- Impurities other than dopants can lead to formation of energy states that lie in the bandgap. These are called as trap states
- The energy location of these trap states play a very important role in their impact on the carrier dynamics.
- Four different process are possible
 - Electron capture
 - Electron emission
 - Hole capture
 - Hole emission
- To boost the recombination the capture of electron and holes should have similar probabilities.



Indirect Recombination

- Let N_t be the recombination trap densities located at energy, E_t .
- The probability of occupation of the trap levels is

$$f(E_t) = f = \frac{1}{1 + e^{\frac{E_t - E_F}{k_B T}}}$$

- The electron capture rate is proportional to the number of electrons in the conduction band ($n = n_0 + \Delta n$) and the number of unoccupied traps $N_t(1 - f)$

$$R_a = \sigma_c \nu_{th} n N_t (1 - f)$$

σ_c is the capture cross-section area of the trap and ν_{th} is the thermal velocity.

Indirect Recombination

- The electron emission rate is proportional to the number of occupied traps

$$R_b = e_n N_t f$$

- Electron emission probability, e_n , depends upon the density of empty states in the conduction band, trap energy w.r.t. to the conduction band.
- For any physical system in thermal equilibrium the capture and the emission rates must be the same.

$$e_n = \sigma_{cn} \nu_{th} n_i e^{\frac{E_t - E_i}{k_B T}}$$

where we have assumed that the electron concentration, n , can be correctly described by the Maxwell-Boltzmann distribution

Indirect Recombination

- Similarly for holes for hole capture and emission process

$$R_c = \sigma_{cp} \nu_{th} p N_t f$$

$$R_d = e_p N_t (1 - f)$$

$$e_p = \sigma_{cp} \nu_{th} n_i e^{\frac{E_i - E_t}{k_B T}}$$

- Consider a system where in semiconductor block is illuminated with a light which increases the electron concentration more than its equilibrium value. And this increases the capture rate, R_a . In steady state we can write

$$G_L + R_b - R_a = 0$$

- Similarly for holes as well we can write

$$G_L + R_d - R_c = 0$$

Indirect Recombination

- This allows us to write

$$R_a - R_b = R_c - R_d \equiv U$$

- Combining different rates we get

$$f = \frac{\sigma_{cn}n - \sigma_{cp}p_1}{\sigma_{cn}(n + n_1) - \sigma_{cp}(p + p_1)}$$

- This allows us to write

$$U = \frac{\sigma_{cp}\sigma_{cn}\nu_{th}N_t(pn - n_i^2)}{\sigma_{cn}(n + n_1) + \sigma_{cp}(p + p_1)}$$

where

$$n_1 = n_i e^{\frac{E_t - E_i}{k_B T}} \quad p_1 = n_i e^{\frac{E_i - E_t}{k_B T}}$$

- Further rearranging we get

$$U = \frac{(pn - n_i^2)}{\tau_{p0}(n + n_1) + \tau_{n0}(p + p_1)}$$
$$\tau_{p0} = \frac{1}{\sigma_{cp}\nu_{th}N_t} \quad \tau_{n0} = \frac{1}{\sigma_{cn}\nu_{th}N_t}$$

- Substituting the values of n_1 and p_1 and assuming $\tau_{n0} = \tau_{p0} = \tau_0$

$$U = \frac{(pn - n_i^2)}{\tau_0 \left(n + p + 2n_i \cosh\left(\frac{E_t - E_i}{k_B T}\right) \right)}$$

Indirect Recombination – Limiting case

- If we assume that there is a low level injection ($\Delta n = \Delta p < n_0$)

$$\tau = \Delta n \left(\frac{\tau_{p0}(n + n_1) + \tau_{n0}(p + p_1)}{pn - n_i^2} \right) \approx \left(\frac{\tau_{p0}(n + n_1) + \tau_{n0}(p + p_1)}{n_0} \right)$$

- In an n-type semiconductor the excess carrier life time can be approximated as

$$\tau \approx \tau_{p0} = \frac{1}{\sigma_{cp}\nu_{th}N_t}$$

- Surprisingly the excess carrier life time in an n-type semiconductor depends on the parameters of the hole
- Similarly in a p-type semiconductor the excess carrier life time can be approximated as

$$\tau \approx \tau_{n0} = \frac{1}{\sigma_{cn}\nu_{th}N_t}$$

Indirect Recombination – Limiting case

- In the case of high level injection

$$\tau \approx \tau_{p0} + \tau_{n0}$$

- Experimentally, it has been found that in silicon and germanium $\tau_{n0} \geq \tau_{p0}$
- The indirect recombination is also called as Shockley-Reed-Hall (SRH) recombination process.
- This is the dominant recombination process in Si and Ge.

Auger Recombination – Band to Band Process

- Auger recombination is 3 particle process i.e. an 2 electrons and one hole or 2 holes and 1 electron interact
- It is reasonable to write the recombination rate

$$R_A = C_{nnp}n^2p + C_{ppn}p^2n$$

- By the principle of detailed balance, the generation rate in thermal equilibrium can be written as

$$\begin{aligned} G_A &= C_{nnp}n_0^2p_0 + C_{ppn}p_0^2n_0 \\ &= (C_{nnp}n_0 + C_{ppn}p_0)n_i^2 \end{aligned}$$

- So the net recombination rate is

$$U_A = C_{nnp} (n^2p - n_i^2n_0) + C_{ppn} (p^2n - n_i^2p_0)$$

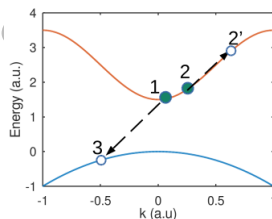


Illustration of 2 electrons and 1 hole process

Auger Recombination – Band to Band Process

- This allows us to write the recombination rate as

$$\begin{aligned}\tau_A &= \frac{\Delta n}{U_A} \\ &= \frac{1}{C_{nnp} (2n_i^2 + p_0^2) + C_{ppn} (2n_i^2 + p_0^2)}\end{aligned}$$

- Generally dominant only in the presence of very high carrier concentrations and in low band gap materials.
- How does this expression simplify for high level and low level injection ?

Putting it all together

- Assuming that all the process are independent of each other

$$\frac{1}{\tau} = \frac{1}{\tau_R} + \frac{1}{\tau_P} + \frac{1}{\tau_A}$$

where τ_R is the excess carrier life time due to radiative recombination
 τ_{SRH} is the excess carrier life time due to SRH recombination process
 τ_A is the excess carrier life time due to Auger recombination process.

- The net excess carrier lifetime would naturally be added reciprocally.

Carrier Continuity

- Nature is very good in book keeping
 - Water cycle example
- Mathematically writing it

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + G - R = \frac{1}{q} \frac{\partial J_n}{\partial x} - \frac{\Delta n}{\tau_n}$$

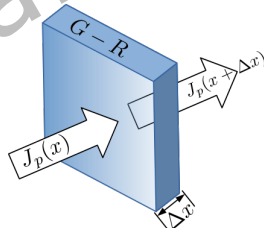
- Similarly for holes as well

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} + G - R = -\frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{\Delta p}{\tau_p}$$

- J_n and J_p are the electron and hole current respectively

$$J_n = q\mu_n n \mathcal{E} + qD_n \frac{dn}{dx} \quad J_p = q\mu_p p \mathcal{E} - qD_p \frac{dp}{dx}$$

- Generally we are interested in steady state.



Diffusion Length

- In a steady state the time derivative is zero. So this allows us to write the continuity equations as

$$\frac{1}{q} \frac{\partial J_n}{\partial x} - \frac{\Delta n}{\tau_n} = 0 \quad - \frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{\Delta p}{\tau_p} = 0$$

- Let us further assume that the drift component is zero. So substituting the diffusion component in place $J_{n/p}$ of we get

$$\frac{\partial^2 \Delta n}{\partial x^2} = \frac{\Delta n}{D_n \tau_n} = \frac{\Delta n}{L_n^2}$$
$$\frac{\partial^2 \Delta p}{\partial x^2} = \frac{\Delta p}{D_p \tau_p} = \frac{\Delta p}{L_p^2}$$

- L_n and L_p are the diffusion lengths for electrons and holes respectively

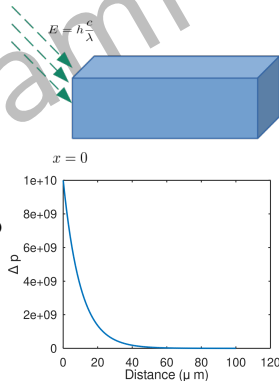
Diffusion Length

- Consider a piece of semiconductor in which we generate an excess of holes at $x = 0$.
- The solution to the above equations can be written as

$$\Delta p(x) = \Delta p(0)e^{-\frac{x}{L_p}}$$

- Diffusion length represents the distance at which the excess carrier concentration drops to $\frac{1}{e}$ value of at its source
- The probability that the excess hole generates diffuses up to 'x' without recombining is

$$\frac{\Delta p(x)}{\Delta p(0)} = e^{-\frac{x}{L_p}}$$



Diffusion Length

- The probability that the excess carrier will recombine in dx distance is

$$\frac{\Delta p(x) - \Delta p(x + dx)}{\Delta p(x)} = \frac{1}{L_p} dx$$

- Thus the probability that the excess carrier travels up to x and then recombines in $x = dx$ is given by

$$P_R(x, x + dx) = e^{-\frac{x}{L_p}} \frac{1}{L_p} dx$$

- Thus the average distance diffused by the excess carrier is

$$\langle x \rangle = \int_0^{\infty} x e^{-\frac{x}{L_p}} \frac{1}{L_p} dx = L_p$$

- Thus the Diffusion length is the average distance the a carrier diffuses without recombining

Poisson's equation

- Gauss's law states that electric flux coming out of a volume is equal to the net charge enclosed by the volume

$$\int_S \mathbf{D} \cdot d\mathbf{S} = \int_{\Omega} \rho dV$$
$$\Rightarrow \int_{\Omega} \nabla \cdot \mathbf{D} dV = \int_{\Omega} \rho dV$$

where $\mathbf{D}$ is the electric flux density, ρ is the net charge enclosed in the volume, Ω

- The flux density is related to the $\mathcal{E}$ through the permittivity of the material

$$\mathbf{D} = \epsilon \mathcal{E} = -\epsilon \nabla V$$

$$\nabla \cdot \epsilon \nabla V = -q (N_D - N_A + p - n)$$

How does a simulator work ?

